

Machine Learning Fundamentals

Forms of Learning

Supervised, unsupervised, semi-supervised, reinforcement learning

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Outline

- 1 Introduction to Machine Learning
- 2 Supervised Learning
- 3 Unsupervised Learning
- 4 Semi-Supervised Learning
- 5 Reinforcement Learning
- 6 Comparison and Summary

What is Machine Learning?

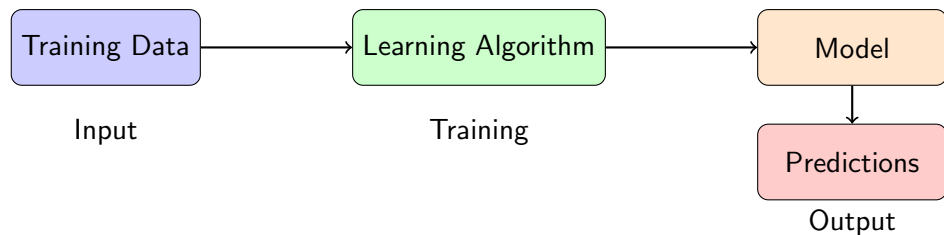
Definition

Machine Learning is a subset of Artificial Intelligence that enables systems to learn and improve from experience without being explicitly programmed.

- Focuses on developing algorithms that can access data and learn from it
- Builds models based on sample data (training data)
- Makes predictions or decisions without human intervention
- Applications: spam filtering, image recognition, recommendation systems, autonomous vehicles

Reference: Russell & Norvig (2020), Chapter 18

Machine Learning Paradigm



The learning process involves extracting patterns from data to create a model that can generalize to unseen examples.

Forms of Learning

Machine Learning can be categorized into four main paradigms:

① Supervised Learning

- Learning from labeled data
- Teacher provides correct answers

② Unsupervised Learning

- Learning from unlabeled data
- Discovery of hidden patterns

③ Semi-Supervised Learning

- Combination of labeled and unlabeled data
- Leverages both paradigms

④ Reinforcement Learning

- Learning through interaction with environment
- Reward-based optimization

Reference: Bishop (2006), Chapter 1

Supervised Learning: Overview

Definition

Supervised learning involves training a model on labeled data where each training example consists of an input and its corresponding desired output.

Key Characteristics:

- Training set: $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
- Goal: Learn function $f : X \rightarrow Y$ that maps inputs to outputs
- Supervision provided by labeled examples
- Performance evaluated on test data

$$\text{Minimize: } \frac{1}{n} \sum_{i=1}^n L(f(x_i), y_i)$$

Reference: Russell & Norvig (2020), Section 18.6

Types of Supervised Learning

1. Classification

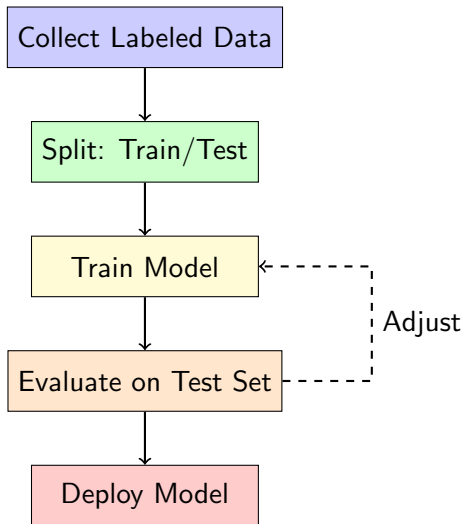
- Discrete output values
- Predicts class labels
- Examples:
 - Email: spam/not spam
 - Image: cat/dog/bird
 - Medical: disease/healthy
- Algorithms: Decision Trees, SVM, Neural Networks

2. Regression

- Continuous output values
- Predicts numerical values
- Examples:
 - House price prediction
 - Stock market forecasting
 - Temperature estimation
- Algorithms: Linear Regression, Polynomial Regression

Reference: Bishop (2006), Chapters 4-5

Supervised Learning Process



Supervised Learning: Advantages and Challenges

Advantages:

- Clear objective and evaluation
- Strong theoretical foundation
- Well-established algorithms
- Directly optimizes for task
- Interpretable results

Challenges:

- Requires labeled data
- Labeling is expensive and time-consuming
- May not generalize well
- Sensitive to label quality
- Overfitting risk

Important

The quality and quantity of labeled training data significantly impacts model performance.

Reference: Rich et al. (2009), Chapter 17

Unsupervised Learning: Overview

Definition

Unsupervised learning involves finding hidden patterns or structures in unlabeled data without explicit guidance on what to predict.

Key Characteristics:

- Training set: $\{x_1, x_2, \dots, x_n\}$ (no labels)
- Goal: Discover underlying structure in data
- No supervision or feedback
- Exploratory data analysis

$$\text{Find: } \theta^* = \arg \max_{\theta} p(X|\theta)$$

Reference: Bishop (2006), Chapters 9, 12

Types of Unsupervised Learning

1. Clustering

- Group similar data points together
- Examples: K-means, Hierarchical clustering, DBSCAN
- Applications: Customer segmentation, document organization

2. Dimensionality Reduction

- Reduce number of features while preserving information
- Examples: PCA, t-SNE, Autoencoders
- Applications: Data visualization, feature extraction

3. Anomaly Detection

- Identify unusual patterns or outliers
- Examples: One-class SVM, Isolation Forest
- Applications: Fraud detection, system monitoring

Reference: Russell & Norvig (2020), Section 18.10

Clustering Example: K-means Algorithm

Algorithm 1 K-means Clustering

- 1: Initialize k cluster centroids randomly: $\mu_1, \mu_2, \dots, \mu_k$
 - 2: **repeat**
 - 3: **Assignment step:** Assign each point to nearest centroid
 - 4: **for** each data point x_i **do**
 - 5: $c_i \leftarrow \arg \min_j ||x_i - \mu_j||^2$
 - 6: **end for**
 - 7: **Update step:** Recompute centroids
 - 8: **for** each cluster j **do**
 - 9: $\mu_j \leftarrow \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$
 - 10: **end for**
 - 11: **until** convergence or max iterations reached
-

Reference: Bishop (2006), Section 9.1

Unsupervised Learning: Advantages and Challenges

Advantages:

- No need for labeled data
- Discovers unknown patterns
- Useful for exploration
- Scalable to large datasets
- Preprocessing for supervised learning

Challenges:

- Evaluation is difficult
- Results may be ambiguous
- Requires domain knowledge for interpretation
- No clear objective function
- May find spurious patterns

Note

Unsupervised learning is particularly valuable when labeled data is scarce or expensive to obtain.

Reference: Rich et al. (2009), Chapter 17

Semi-Supervised Learning: Overview

Definition

Semi-supervised learning uses a combination of small amounts of labeled data and large amounts of unlabeled data to train models.

Key Characteristics:

- Training set: $\{(x_1, y_1), \dots, (x_l, y_l)\}$ (labeled) + $\{x_{l+1}, \dots, x_{l+u}\}$ (unlabeled)
- Typically: $l \ll u$ (few labeled, many unlabeled)
- Leverages structure in unlabeled data
- Bridges supervised and unsupervised learning

Motivation: Labeling is expensive, but unlabeled data is abundant

Reference: Russell & Norvig (2020), Section 18.6.3

Semi-Supervised Learning Approaches

1. Self-Training

- Train on labeled data, predict on unlabeled data
- Add confident predictions to training set
- Iteratively retrain the model

2. Co-Training

- Use multiple views of the data
- Train separate classifiers on different feature sets
- Classifiers teach each other

3. Graph-Based Methods

- Represent data as graph with nodes (samples) and edges (similarity)
- Propagate labels through graph structure
- Smooth labels over similar instances

Reference: Bishop (2006), Section 14.1

Semi-Supervised Learning: Assumptions

Semi-supervised learning relies on key assumptions about data structure:

1 Smoothness Assumption

- Points close to each other likely share same label
- Decision boundary should lie in low-density regions

2 Cluster Assumption

- Data forms discrete clusters
- Points in same cluster likely share same label

3 Manifold Assumption

- High-dimensional data lies on lower-dimensional manifold
- Similar points on manifold have similar labels

Reference: Russell & Norvig (2020), Section 18.6

Semi-Supervised Learning: Benefits and Applications

Benefits:

- Reduces labeling cost
- Improves model performance
- Leverages abundant unlabeled data
- Useful when labels are expensive
- Often outperforms supervised learning with limited labels

Applications:

- Text classification
- Web content classification
- Speech recognition
- Medical diagnosis
- Image classification
- Protein sequence analysis

Key Insight

Semi-supervised learning is most effective when unlabeled data provides meaningful information about the underlying structure relevant to the prediction task.

Reference: Rich et al. (2009), Chapter 17

Reinforcement Learning: Overview

Definition

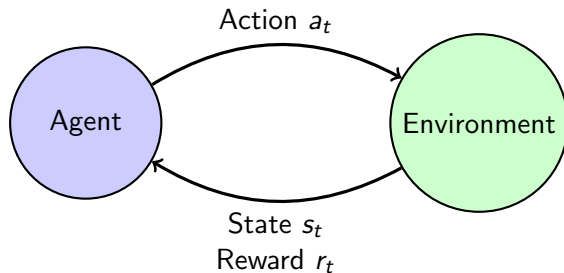
Reinforcement Learning involves an agent learning to make decisions by interacting with an environment to maximize cumulative reward over time.

Key Components:

- **Agent:** The learner or decision maker
- **Environment:** What the agent interacts with
- **State (s):** Current situation of the agent
- **Action (a):** What the agent can do
- **Reward (r):** Feedback from environment
- **Policy (π):** Strategy for selecting actions

Reference: Russell & Norvig (2020), Chapter 22

The Reinforcement Learning Framework



Interaction Cycle:

- 1 Agent observes state s_t
- 2 Agent selects action a_t based on policy π
- 3 Environment transitions to new state s_{t+1}
- 4 Agent receives reward r_t
- 5 Process repeats

Reference: Russell & Norvig (2020), Section 22.1

Key Concepts in Reinforcement Learning

1. Markov Decision Process (MDP)

- Formal framework: (S, A, P, R, γ)
- States, Actions, Transition probabilities, Rewards, Discount factor

2. Value Function

$$V^{\pi}(s) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

Expected cumulative discounted reward from state s

3. Q-Function (Action-Value Function)

$$Q^{\pi}(s, a) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

Expected return from taking action a in state s

Reference: Russell & Norvig (2020), Section 22.2

Reinforcement Learning Algorithms

1. Value-Based Methods

- Q-Learning: Learn optimal action-value function
- SARSA: On-policy TD control
- Deep Q-Networks (DQN): Neural network approximation

2. Policy-Based Methods

- Policy Gradient: Directly optimize policy
- REINFORCE algorithm
- Actor-Critic methods

3. Model-Based Methods

- Learn model of environment
- Plan using the learned model
- Examples: Dyna-Q, AlphaZero

Reference: Russell & Norvig (2020), Sections 22.3-22.5

Exploration vs. Exploitation

A fundamental challenge in reinforcement learning:

Exploitation:

- Use current knowledge
- Choose best known action
- Maximize immediate reward
- Risk missing better options

Exploration:

- Gather new information
- Try unknown actions
- May sacrifice immediate reward
- Potential for better long-term rewards

Common Strategies:

- ϵ -greedy: Explore with probability ϵ
- Softmax: Probabilistic action selection
- Upper Confidence Bound (UCB)
- Thompson Sampling

Reference: Russell & Norvig (2020), Section 22.2.3

Reinforcement Learning: Applications

Successful Applications:

- **Game Playing:** AlphaGo, Chess engines, Atari games
- **Robotics:** Robot manipulation, locomotion, navigation
- **Autonomous Vehicles:** Self-driving cars, drones
- **Resource Management:** Data center cooling, energy optimization
- **Finance:** Algorithmic trading, portfolio management
- **Healthcare:** Treatment optimization, drug dosing
- **Recommendation Systems:** Personalized content delivery
- **Natural Language:** Dialogue systems, machine translation

Breakthrough

RL has achieved superhuman performance in complex domains like Go and StarCraft II.

Reference: Rich et al. (2009), Chapter 17

Comparison of Learning Paradigms

Type	Data	Feedback	Goal
Supervised	Labeled (x, y)	Direct labels	Learn input-output mapping
Unsupervised	Unlabeled (x)	None	Discover structure
Semi-supervised	Mostly unlabeled, few labeled	Partial labels	Improve with limited labels
Reinforcement	State-action sequences	Rewards	Learn optimal policy

Key Distinction: The type and availability of feedback determines which learning paradigm to use.

Reference: Bishop (2006), Chapter 1

Machine Learning Forms:

- **Supervised Learning:** Learn from labeled examples for classification and regression
 - Pros: Clear objectives, strong performance
 - Cons: Requires expensive labeled data
- **Unsupervised Learning:** Discover patterns in unlabeled data
 - Pros: No labeling needed, exploratory
 - Cons: Difficult evaluation, ambiguous results
- **Semi-Supervised Learning:** Leverage both labeled and unlabeled data
 - Pros: Reduces labeling cost, improves performance
 - Cons: Requires structural assumptions
- **Reinforcement Learning:** Learn through interaction and rewards
 - Pros: Handles sequential decisions, achieves superhuman performance
 - Cons: Sample inefficient, requires extensive exploration

-  Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
-  Rich, E., Knight, K., & Nair, S. B. (2009). *Artificial Intelligence* (3rd ed.). McGraw-Hill.
-  Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

Thank You!

Questions?